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ORIGINAL ARTICLE



Deadbeat predictive direct power control of interleaved buck converter-based fast battery chargers for electric vehicles

Hyeon-Ho Choi¹ · Seok-Hyeon Kim¹ · Jeon-Ho Kim² · Young-Hoon Choi² · Hyeon Beom Lee²

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Abstract

This paper presents a deadbeat predictive direct power control (DB-DPC) scheme for a fast battery charger topology for dynamic performance and stability. Recently, battery chargers have been adopting parallel multi-cell configuration based on efficient multi-cell devices to increase power capacity, which reduces the charging duration. However, the inherent instability of the battery is expected to occur during the fast charging period. Therefore, the proposed DB-DPC provides a stable steady state when it includes real-time estimation of the variable battery impedance. Furthermore, the DB-DPC effectively reduces the ringing and settling time, which demonstrates fast dynamic response characteristics when compared to proportional–integral (PI) control. The control performance is verified by various simulation and experimental results.

Keywords Deadbeat predictive control · Direct power control · Multi-cell interleaved converter · Fast power settling control · Electric vehicle charger

1 Introduction

Parallel multi-cell converters have been widely used in power conversion systems that require high-power capacity and high efficiency such as power factor compensation devices, semiconductor power systems, and even recently fast battery chargers for electric vehicles (EVs) [1–5]. Figure 1 shows a buck converter with two parallel legs for fast battery chargers. All four of the high-side of the same components including metal oxide semiconductor field-effect transistors (MOSFETs), diodes, and capacitors. All of the legs conduct their switching operations at the same switching frequency.

There are two types of power semiconductor devices: the discrete type and packaged module type. The packaged module type devices are designed with multiple existing devices or an integrated circuit topology in a package. For example, there are air-pack modules that include three half-bridge legs or a buck converter module that includes one

MOSFET and one diode. Generally, module-type power devices provide high-power capability and easy installation due to their packaged construction. However, the price of module-type devices is higher than that of discrete-type devices. The module-type devices are mostly used in power converters that require a small volume and a low cost. In addition, the parallel connection of discrete-type devices is not efficient when it comes to the design of high-power converters. Moreover, parallel multi-leg converters can be operated as an interleaved converter to reduce ripple current. A multi-leg interleaved buck converter is adopted as a good solution for applications with a high power and a high current with a low current ripple [6–10]. In the case of battery chargers, ripple current affects the lifetime of the batteries or capacitors. The use of an interleaved converter effectively reduces the ripple current and thereby increases the power capacity by adding non-switching legs.

Low-voltage (LV) batteries are widely applied in EVs due to their high power capacity and cost, and the charge characteristics. The average density of LV is significantly better than of standard vehicle applications. Despite their several advantages, they are fragile and require a protection circuit for safe charging. Moreover, aging is a concern for most LV batteries. However, many manufacturers rarely affect about this issue. It has been reported that frequent current exchange and large current ripple result

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where a state feedback control law is derived from three regions which are analysed based on some constraints to check feasibility using phase plane concepts. In Karami and Khatami (2007), another method using nonlinear piecewise linear a PI controller to ensure the global stability of a DC-DC converter. There is another approach to achieve the asymptotic stability, by using the nonlinear controller. A geometric based non-linear method, referred to as the boundary control, which is employed to derive the control law to lead the CPL, reach the desired operating point (Karami and Karami (2005)). This technique makes the state variables of the system converge to achieve a boundary, a stable steady-state system. Lyapunov method is used to ensure that the selected boundary was chosen which subject to achieve asymptotic convergence at target CPL operation. A large signal method using Tikhonov approximation the nonlinear DC-DC system with linear system and study the region of attraction-convergence gives equilibrium point (Karami et al. (2007)). A sliding mode delay robust controller (DC/DC) based on properties of differential power system was proposed in Khan et al. (2004), however it didn't prove global stability. A linear analysis method using the boundary control is based on the proposed nonlinear sliding mode control was proposed in Taheri (2017), however stability analysis was conducted based on nonlinear conditions instead of the constant average impedance load. In Zhang et al. (2016) the system of a polytopic uncertainty set is considered and the convex optimization problem focused to find the equilibrium point. However, there is conservatism in their solution. The Lyapunov function with the robustness controller is proposed in Hossain and Hossain (2017), but there is chattering issue because of the the continuous sliding mode controller. It clearly shows that proposed method for different CPLs. Characteristic modeling for multiple CPLs and stability with closed impedance was proposed in Chongchai Chaisri et al. (2015). When the parameters such as inductance and capacitance of the converter are equal or close to a certain range, disruptions will lose stability because of the resonance, appropriate design efforts to control the power system including CPL, by writing a function mainly to determine stability range (Song et al. (2017)). However it was applied around the small signal operating point. Hybrid control controller with a proportional integral (PI) algorithm to regulate the output voltage of the converter was proposed in Shams et al. (2008). A proposed nonlinear feedback loop which is called a loop cancellation technique has been developed in M. Karami et al. (2010), where the loop is placed in parallel with the classical voltage feedback path to achieve fast steady state error and damping at the same time. A nonlinear controller was proposed in Nabil Mohamed et al. (2009) to control a permanent magnet synchronous motor (PMSM) based on adaptive control method. This large signal analysis proved that it is very helpful in understanding the behavior of CPLs. It helps

to control the system that has large changes caused upon any point caused by conversion or disconnection of large load or generation. A robust impedance based stability criterion method was proposed for grid-connected converter in Yu (2017), where it checks stability of global when frequency of the partial system. If global when frequency goes above Nyquist stability criterion, then system is stable. It is very simple method and doesn't need error calculation when the system complexity increases, but it suffers from instability when grid impedance is number of connected location have been increased above certain limit. This cascaded converter system (Mansell (2013)), any change in downstream converter caused disturbance in the upstream converter but there is a time delay in its feedback loop that degrades system stability as an active feedback loop was transferred from the downstream to the upstream converter stability. A nonlinear adaptive controller was proposed to stabilize DC-DC converter, the proposed model reference adaptive controller with oscillation with high frequency signals (Fu (2016)). Power circuit control was proposed in (Johansson (2017)) which is consisted of primary controller based on sliding approach to ensure stability with CPL, and integral secondary controller to correct voltage fluctuations. However, only local asymptotic stability was demonstrated it didn't ensure global asymptotic stability. Quadratic Lyapunov function was proposed for DC-DC with CPL, to stabilize the system but suffers from rate separation with increasing CPLs (Hafmann et al. (2015)). Energy buffer power converter for a constant power LED lighting load that presents a variable load impedance to the electrical system has been proposed in Li et al. (2017). State feedback linearization method was also proposed in General (2015). A lot of the previous methods don't ensure global asymptotic stability and to achieve controller parameter uncertainties which are the motivation for this work.

1.2 State Contributions

Based on the above mentioned results and shortcomings, in this work we propose a simple and robust PI controller based on Karami and Khatami theorem. This controller is not only able to mitigate the instability issue caused by the CPL nonlinearity but also the uncertainty in system parameters. However, model-based modeling structures have been avoided by designing a robust PI controller for model model plant which means that it will be able to operate around different operating points. This can be achieved based on Karami and Khatami theorem is that the PI model has robust (Fu et al. (2002)). Although Nyquist plot and root locus were used to stabilize the system but these methods are graphical and not a provide analytical characteristics with a stabilizing controller gain. We used the new Karami and Khatami theorem to design a proposed polytopic stabilization the model model

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New high-efficiency multifunctional buck/boost/buckboost PFC converter

Lixiaonan Li^{1,2*} · Chang Yu² · Qianhuiang Chen² · Jun Li² · Shi Lin^{2,3}

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Abstract

A new three-phase buck/boost/buckboost power factor correction rectifier is presented in this paper. In the converter, only one high-frequency switching action exists for each sample period, and only one inductor is used. As a result, system efficiency can be improved. The converter topologies are analyzed for the three modes, the control design of the rectifier is discussed, and the output duty is adjusted between switches according to different intervals of input voltage. Experiments are conducted to verify the effectiveness of the proposed rectifier.

Keywords Buck PFC · Boost PFC · Buckboost PFC · Conduction loss · Converter

1 Introduction

Charging technology, which is converting alternative grid energy into DC batteries, is attracting more interest with the development of electric vehicles. PFC converters are adopted in the first stage of the charger, and a DC/DC converter is often used for the second stage. This system with a large capacity, a three-phase PFC is commonly used because the maximum power of single-phase is limited [1, 2]. Theory and methods for reducing switching losses

There are two kinds of PFC rectifiers: boost-type and buck-type. The former [3] is single and adapted rectifier but requires the output voltage to be higher than the input peak line voltage. In several cases requiring a low DC output voltage, such as the charger for electric vehicles, a buck-type DC/DC converter should be connected serially to the boost-type PFC. This kind of system design is very simple and direct, and has become sophisticated. Many academic or engineering achievements exist for this type of converter,

which include the control of output voltage, input current, FCM modulation, or system efficiency improvement [4–7].

The buck-type PFC has become an engaging research focus because power factor is not greatly improved, conversion efficiency, while the converter has become more complicated in system-parameter topology and improves system efficiency at power density. In [8], three single-phase PFC converters are combined to form a three-phase buck-type rectifier. The system behavior curves can be properly determined in the three phases by proper modulation such that the system forms the three-phase currents in the current through a late pass flow. In [9], a novel buck-type PFC rectifier is proposed, where the modulation, control, converter design, and modulation strategy are discussed in detail. The key of the topology is the control of two high-frequency switches and three couples of serially anti-connected IGBTs. More research results are presented in [10–13].

A new multifunctional buck/boost/buckboost PFC converter is proposed in this paper, because it with output voltage range may be applied to several cases. The output voltage can be lower or higher than the input voltage, while the input current can be reduced sinusoidal, and a soft power factor can be achieved. Compared with multifunctional three-phase/boost-type PFC combined with a buck-type DC/DC converter, the proposed one is simply a single-stage topology. Compared with the buck-type PFC, the proposed one can achieve a higher output voltage.

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